MEASUREMENT AND DATA

Units and Symbols

For almost all of the work done in Years 11 and 12, the SI (Système International) system of units will be used. There are just seven ***fundamental quantities*** that form the basis of all measurements in science.

table 1.1 The seven fundamental quantities of the SI system

Unit	Symbol	Physical quantity
metre	m	distance
second	s	time
kilogram	kg	mass
ampere	A	electrical current
kelvin	K	thermodynamic temperature
mole	mol	amount of substance
candela	cd	luminous intensity

A ***derived unit*** is a combination of fundamental units.

e.g. A derived unit is the unit of velocity: metre per second. It is written $m\ s^{-1}$.

table 1.2 Some examples of the use of symbols for derived units

Preferred	Correct also		WRONG!	
$m\ s^{-2}$	$m.s^{-2}$	m/s^2	ms^{-2}	
kW h	kW.h		kWh	k Wh
$kg\ m^{-3}$	kg/m^3	$kg.m^{-3}$	kgm^{-3}	
µm			µ m	
N m	N.m		Nm	

Scientific Notation

Measurements are often written in scientific notation. Quantities are written as a number between one and ten and then multiplied by an appropriate power of ten.

(Note that'scientific notation', 'standard notation' and 'standard form' all have the same meaning.)

e.g. Some measurements written in scientific notation are:

$0.054\ m = 5.4 \times 10^{-2}\ m$
$245.7\ J = 2.457 \times 10^{2}\ J$
$2080\ N = 2.080 \times 10^{3}\ N$

You should be routinely using scientific notation to express numbers. An important reason for using scientific notation is that it removes ambiguity about the precision of some measurements.

e.g. A measurement recorded as 240 m could be a measurement to the nearest metre.

$\therefore$ Use 2.40×10^2 m.

The measurement may be to the nearest 10 m.

$\therefore$ Use 2.4×10^2 m.

Prefixes and Conversion Factors

You should be familiar with the prefixes and conversion factors in the table below. The table gives all conversions as a ***multiplying factor***.

Simply multiply the given number by the factor.

table 1.3 Prefixes and conversion factors

Multiplying factor		Prefix	Symbol
1 000 000 000 000	10^{12}	tera	T
1 000 000 000	10^{9}	giga	G
1 000 000	10^{6}	mega	M
1 000	10^{3}	kilo	k
0.01	10^{-2}	centi	c
0.001	10^{-3}	milli	m
0.000 001	10^{-6}	micro	μ
0.000 000 001	10^{-9}	nano	n
0.000 000 000 001	10^{-12}	pico	p

e.g. 2.45×10^3 mm = $2.45 \times 10^3 \times 10^{-3} = 2.45$ m

2.45×10^3 nm = $2.45 \times 10^3 \times 10^{-9} = 2.45 \times 10^{-6}$ m

1.25×10^3 mm^2 = $1.25 \times 10^3 \times (10^{-3})^2 = 1.25 \times 10^{-3}$ m^2

When calculating an area conversion, the factor is "squared".
For a volume conversion, the factor is "cubed".

To convert km h-1 to m s-1, divide by 3.60.

e.g. 70.0 km h^{-1} = $70.0 \div 3.60$ = 19.4 m s^{-1}

Uncertainty in Measurements

All measurements have some amount of uncertainty.

The uncertainty is one half of the finest scale division on the measuring instrument.

e.g. A measuring wheel has an uncertainty of 0.5 m (if it measures to the nearest metre).

A metre rule has an uncertainty of 0.5 mm (smallest division is 1.0 mm)

An electronic balance set to measure grams to two decimal places has an uncertainty of 0.005 g.

Sometimes this uncertainty is referred to as error. ***It is not error***, in that it is not a mistake or something wrong. All measuring instruments have limited precision and the uncertainty is half of the smallest scale division on the instrument.

The uncertainty gives us the range in which a measurement falls. If we measured the length of a stick with a metre rule then we would get a measurement 'plus or minus' half a millimetre.

e.g. Any stick between 127.5 and 128.5 mm long would be measured as 128 mm to the nearest millimetre. We would record this as ***128 ± 0.5 mm***.

The best judgement you can definitely claim is one half of a scale division - it is too difficult to judge to a finer degree. The uncertainty we will still assume, however, is a full half scale division.

e.g. You might measure another stick, one that has a length somewhere between 154 and 155 mm, as ***154.5 ± 0.5 mm***.

There are two ways to record uncertainties.

Absolute uncertainty: include the actual uncertainty in the measurement.

e.g. 154.5 ± 0.5 mm

Percentage uncertainty: expressed as a percentage of the measurement.

e.g. % uncertainty = 0.5/154.5 x 100% = 0.3%

∴ 154.5 ± 0.3 %

Uncertainty in Calculations

An experiment or a measurement exercise is not complete until the uncertainties have been analysed. The report should include an estimate of the total uncertainty. This gives the reader of the report some idea of your confidence in the result.

The following three processes are used for estimating uncertainty.

- When adding or subtracting data, add the absolute uncertainties.
- When multiplying or dividing data, add the percentage uncertainties.
- When raising data to power n, multiply the percentage uncertainty by n.

e.g. Calculate the area of a rectangle measuring 22.6 ± 0.1 mm (± 0.4 %) by 14.2 ± 0.1 mm (± 0.7%)

$$A = (22.6)(14.2) = 3.91 \times 10^2 \text{ mm}^2 \pm 1.1\%$$
$$= \underline{3.91 \times 10^2 \pm 4 \text{ mm}^2}$$

Significant Figures

Significant figures represent a way of expressing how confident we are in a measurement. The final digit carries an element of uncertainty - it has been rounded off.

e.g. An electronic balance measures a mass as 143.24 g.

This has 5 significant figures. The machine rounds the second decimal off to the nearest number - it is the uncertain number.

Rules To Use

- All non-zero digits are significant.

- Zeroes between non-zero digits are significant.
  e.g. 4003 has 3 s.f.

- Zeroes to the right of a decimal point and following a non-zero digit are significant.
  e.g. 1.102×10^3 has 4 s.f. 0.0105 has 3 s.f.

- All other zeroes are not significant.
  e.g. 500 has 1 s.f.

  If the zeroes are significant, write it in scientific notation - 5.00×10^2 has 3 s.f.

- In calculations, the final result should not contain any more significant figures than the data with the least number of significant figures.

  e.g. $2.45 \times 5.6 \times 103.23 = 1.4 \times 10^3$

  5.6 has 2 s.f. so the answer must have 2 s.f. - it has the least number of s.f.